

Legend

The color coding in the flux diagrams is as follows:

green actions performed using the cconfigurator, sequential

yellow actions performed using the cconfigurator, parallel

blue subroutines, jumps and returns

orange branches

light green branches triggered by events

Tags

config configurator actions

actuator name, value

actuator name, values: loop through values upon multiple occurrence

actuator name, action

detector name, action

wait wait on a Qt signal

condition source, signal

bifurcation source, property

case value, goto—instructions

case smaller-than, goto—instructions

case smaller-than-or-equal, goto—instructions

case larger-than, goto—instructions

case larger-than-or-equal, goto—instructions

default goto—instructions

loop scan definition followed by instruction tags

scan mode

actuator name, values

actuator name, from, to, steps

actuator name, lin-from, lin-to, lin-steps, log-to, points-per-decade[, neg-log-to]

endcondition source, property (bool)

sub name of subroutine

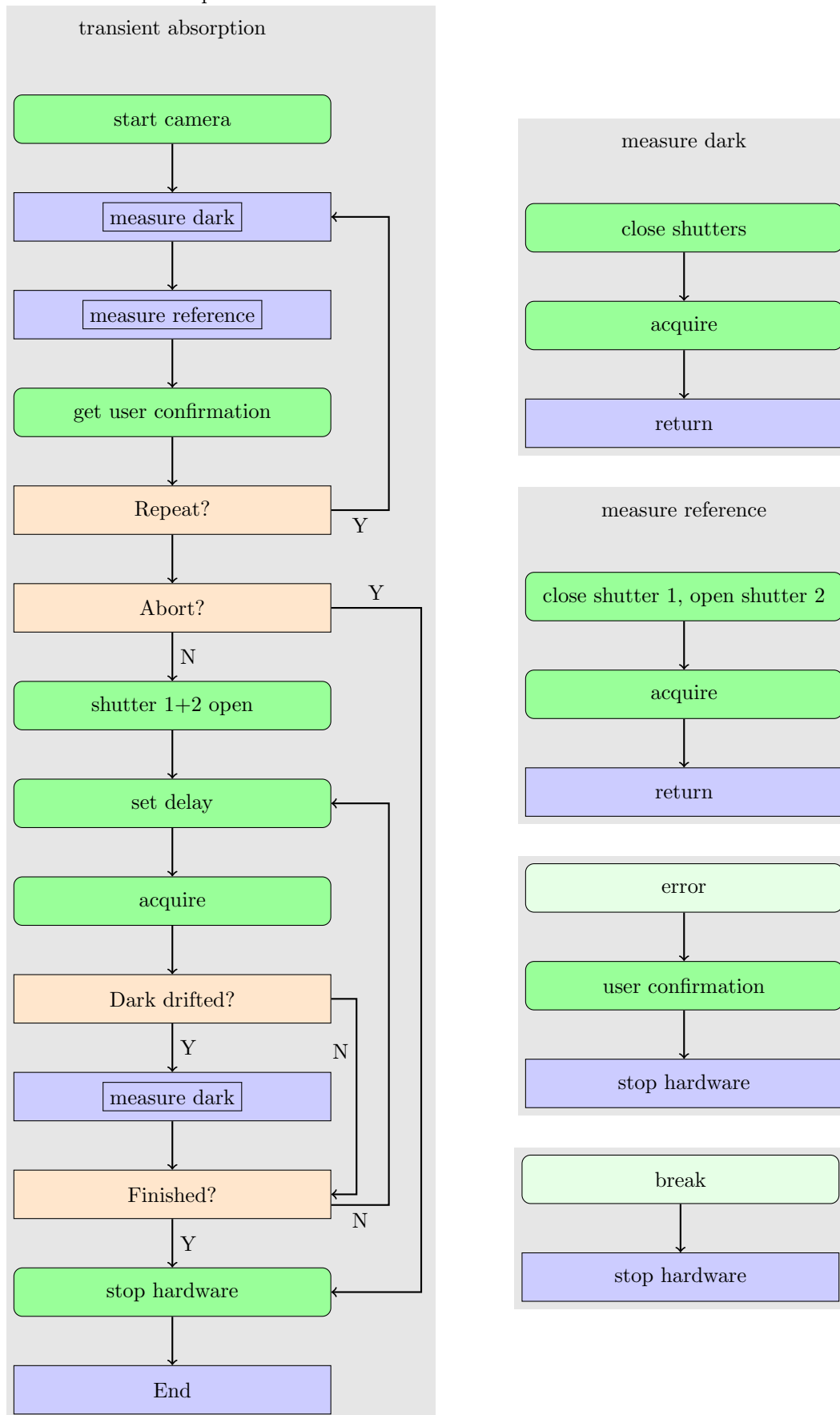
return return from subroutine / event handler

goto name item to jump to

event list of instruction tags entered upon event, ends either with goto or return tag

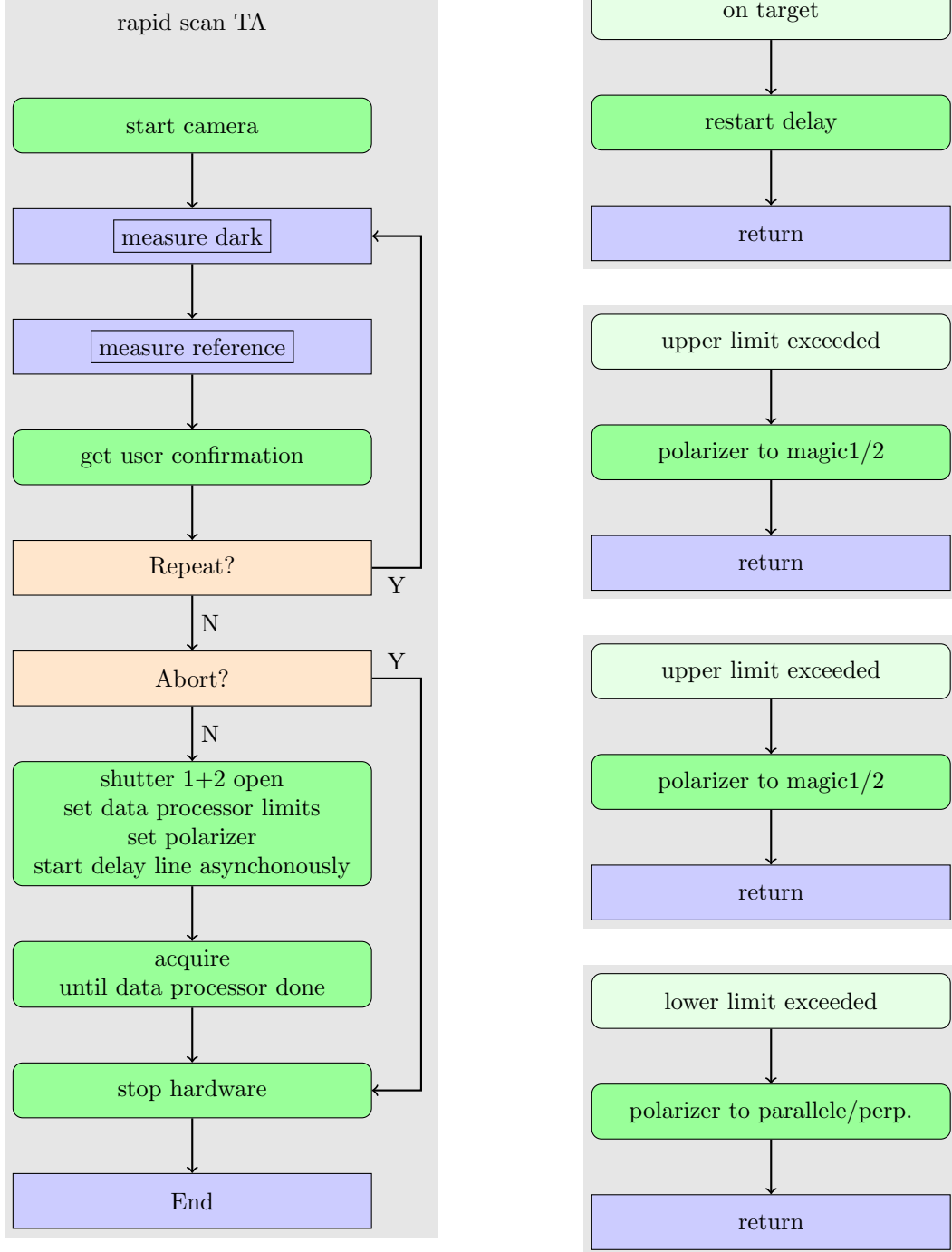
Transient Absorption

Measure absorption spectra as a function of delay between excitation and probe pulse. As preparation measure the cameras' dark signal and the spectrum of the probe light. The former may have to be re-recorded upon thermal drifts.



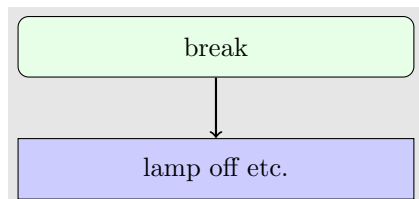
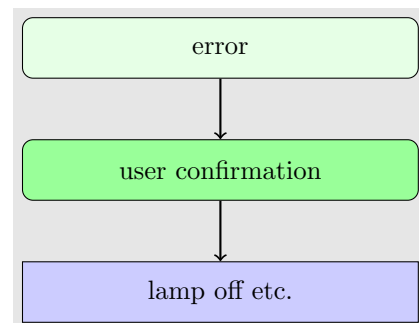
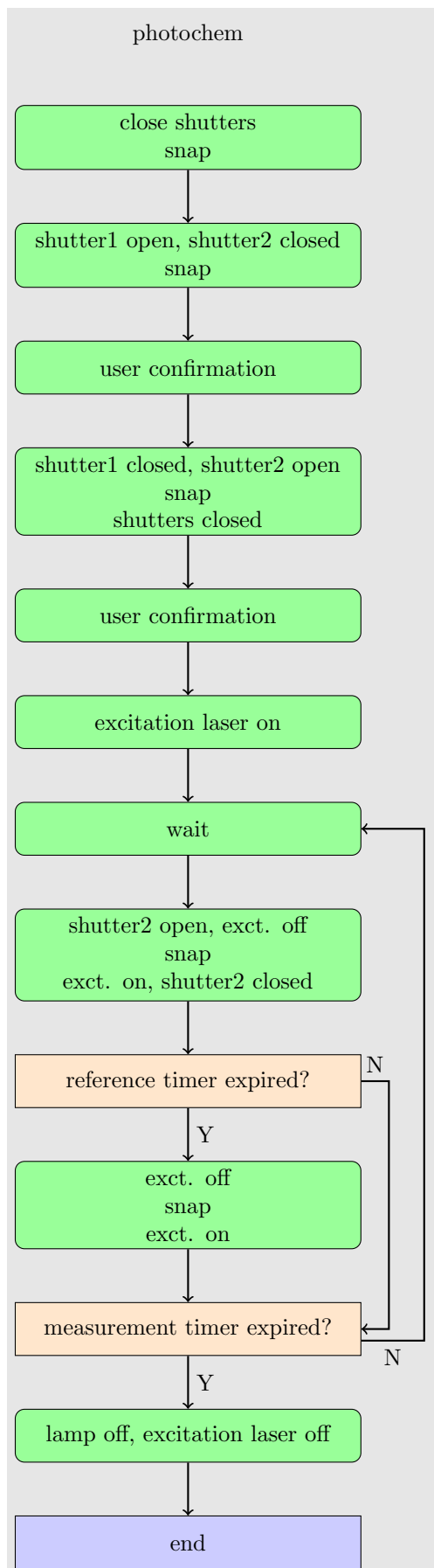
Variant Rapid Scan Transient Absorption

Instead of scanning the time delay in consecutive steps in a step-scan fashion, the optical delay line is rapidly moved forth and back. Its actual position is recorded together with the camera at each trigger from the laser system and the resulting spectra are sorted into the corresponding time bins. Once the target, i.e. the end of the scanned range, has been reached, the actuator notifies the sequencer about this event. By means of an asynchronous or parallel or history state mechanism, the sequencer sends the delay line to the other end of the scan range and continues acquiring. When approaching the target at one end of the scanned range, another signal emitted from the actuator or the data processing unit may be used to notify an end of range and use the remaining stretch to the real end to turn a polarizer or perform any other needed parameter or hardware update.

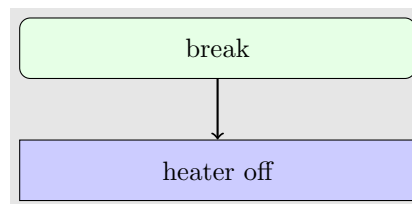
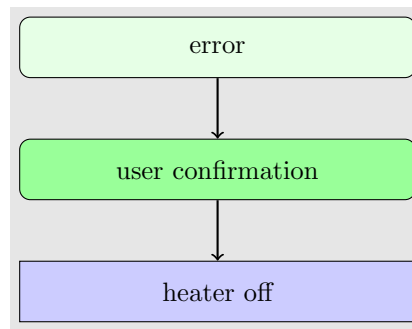
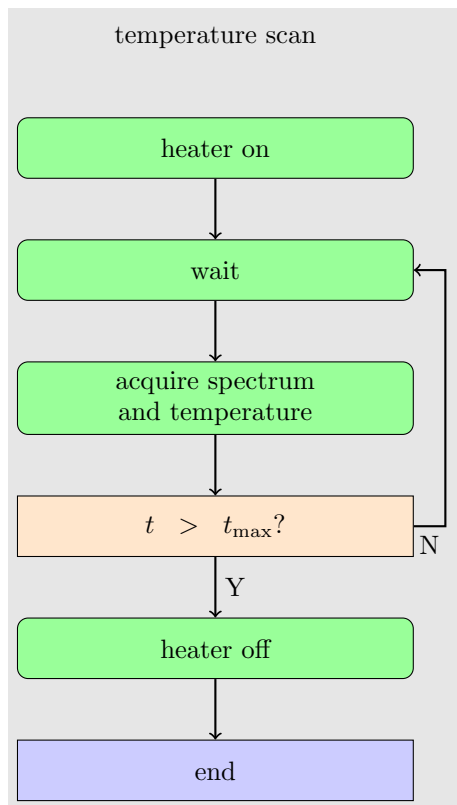


Dark, reference, break and error as with normal transient absorption.

Photochemical Reaction



Temperature Dependent Absorption



Wilson experiemnt

